sor days of February 2021, as shown in Fig. 7(a), 7(b), and 7(c). This is shown by the grey-scale curve in the plots.

What is shockingly observed is that the pre-indications or precursors related to an earthquake and a major volcanic eruption like this Tonga volcano are identical in nature but vary only in time scale. The geophysical process for a volcano can be thought as a slow-motion video where the speed is slower than normal. For an earthquake, the process can be thought as a fast-forward mode where the playback speed is higher than normal.

In other words, the process for a volcano continues for several weeks while that for an earthquake it is wrapped up in just a few hours' time. But the nature of reduction in signal strength of a VLF radio wave is identical and therefore can be successfully used for the prediction of future earthquakes and volcanic eruptions alike.

! Earthquake Alert ! **भूकंपाविषयी महत्वाची सूचना** भूचाल हेतु महत्वपूर्ण सूचना	Number: Date/Time:	11/2021 14-Feb-2021, 1900IST (1330UTC/GMT)
Marathi : (Local)	भूकंपविषयी संशोधन करणाऱ्या उपकरणाकडून प्राप्त माहिती नुसार येत्या १२ दिवसात (**१४ फेब्रुवारी ते २६ फेब्रुवारी २०२१ दरम्यान**) खालील ठिकाणी मोठ्या भूकंपाची शक्यता राहील. भूकंपाची तीव्रता जास्त म्हणजे रिष्टर प्रमाणानुसार **M7** किंवा त्यापेक्षा जास्त राहण्याची शक्यता आहे. त्यामुळे योग्य ती सुरक्षा-काळजी बाळगण्यात यावी. प्रत्यक्ष घटनेच्या कालावधीत वा तीव्रतेत काही फरक पडू शकतो. १) अंदमान आणि निकोबर बेटे (भारत) २) म्यानमार आणि बंगालच्या उपसागराजवळील काही भाग. भारताच्या पूर्वोत्तर राज्यांमध्ये देखील M4 तीव्रतेचे भूकंप होण्याची शक्यता राहील. ३) इंडोनेशियाचे भूकंप प्रवण क्षेत्र, पापुआ न्यू गिनी ४) न्यूझीलंड आणि आसपास चे दक्षिण-प्रशांत महासागरातील देश ५) मालदिव आणि हिंद-महासागर क्षेत्र	
Hindi: (National)	भूचाल हेतु अनुसंधान करनेवाले उपकरणोसे प्राप्त जानकारि के अनुसार आनेवाले १२ दिनोमे निम्न स्थानोमे या आसपास बडे भूचाल (**M7+**) की संभावना रहेगी। यह संभावना **ता. १४ फरबरी से २६ फरबरी २०२१ तक** बनी रहेगी। कृपया सतर्क रहिये| इस कालावधीमे तथा तीव्रता मे आंशिक बदलाव हो सकता है। १) अंदमान एवम निकोबर द्वीप समूह (भारत) २) म्येनमार तथा बंगाल की खाडी, भारत के पूर्वोत्तर राज्य (M4) की संभावना रहेगी| ३) इंडोनेशिया, पापुआ न्यू गिनी ४) न्यूझीलंड तथा पश्चिमी दक्षिणी-प्रशांत महासागरके देश ५) मालदिव के आसपास हिंदमहासागर क्षेत्र।	
English: (Intl.)	Following an earthquake precursor detected in the VLF signal of 18.200kHz on 14-February-2021 since 0830IST till 1800IST, there is a likely chance of **major earthquake** event/s having magnitudes in the range of **M7+** in the following areas in the **next 12 days**. A slight deviation in the days/magnitude is possible. **The likely dates are - 14-Feb-2021 to 26-Feb-2021.** 1) Andaman and Nicobar Islands or nearby region (India) 2) Myanmar and bordering regions in Bay of Bengal. North-Eastern states of India might also experience M4 scale shocks. 3) Indonesian Seismic Zone, Papua New Guinea 4) Western South Pacific including New Zealand 5) Parts of Indian Ocean Region around Maldives	
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Fig. 8: The earthquake prediction alert that was sent by the author on 14th February 2021, stating the possibility of a major M7+ earthquake in next few weeks in the New Zealand and South Pacific region

Therefore, we can easily say that the geophysical processes associated with major earthquakes and volcanic eruptions are identical and generate ionospheric disturbances that reduce the magnitude of a VLF radio wave propagating using the earth-ionosphere-waveguide-mode of propagation. Such clues in future are undoubtedly going to change our understanding about these geophysical processes, and the VLF technique will certainly open a new branch of study for earthquakes and volcanoes.

Importance of these observations and findings

Natural hazards like earthquakes, tsunamis, and volcanoes pose a greater threat to the human life and property and modern infrastructures. We have some recent examples of the December 2004 Sumatra tsunami that hit the Indian and Sri Lankan coasts causing severe casualties. Similarly, the March 2011 Japan tsunami also had devastating impact in Japan, including damage to the Fukushima Nuclear Power Plant. Therefore, knowing about a possible forthcoming mega quake or a volcanic eruption would certainly be useful in minimising loss and damage to the human life and infrastructures.

With the present observations obtained with reference to the January 2022 Tonga volcano almost a month in advance, and the previously reported earthquake prediction of 4th March 2021 using advance precursors detected a few weeks earlier in February 2021, it is clear that the geophysical processes are sending clues and advance information to the ionosphere and such early signatures can be detected using the propagation of VLF signals. EFY

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After the volcano erupted, some detailed information was obtained from the websites of National Centers for Environmental Information of NOAA, USA and Applied Science at NASA and GNS/CRI, New Zealand. The author is thankful to these websites for providing public access information about this Tonga volcano. All copyright information is honoured with thanks.